

LSB IMAGE STEGANOGRAPHY

Project Submitted by :-

Name :- Prathamesh Bhaskar Deore

Batch :- 25044

ID :- 003

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LSB Image Steganography

Abstract:

Steganography is the technique of concealing the existence of communication by embedding information within other data. Various carrier file formats can be used, but digital images are the most commonly used due to their widespread presence on the internet. There are numerous steganography techniques available for hiding information in images; some are more complex than others, each having its own strengths and limitations. Certain applications require complete invisibility of the hidden data, while others focus on embedding larger amounts of information.

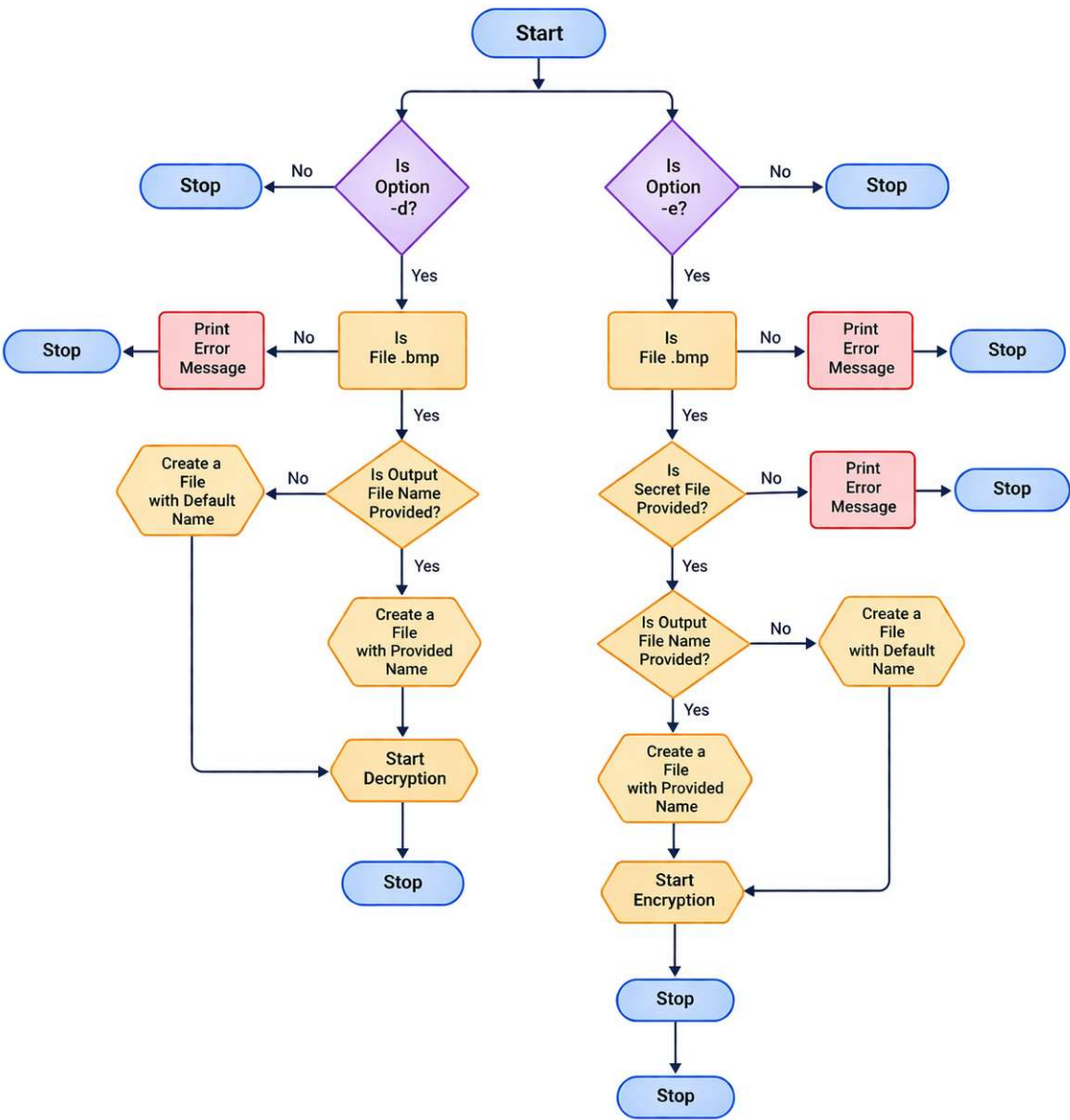
Steganography involves hiding confidential or sensitive data within something that appears ordinary. It is often confused with cryptography, as both are used to secure information. However, the key difference is that steganography hides the existence of the information itself, making it appear as though no data is present. As a result, anyone viewing the carrier file will not suspect or attempt to extract hidden information.

In essence, steganography takes advantage of human perception, as humans are not trained to detect hidden data within files. Although tools exist to perform steganographic analysis, the most common approach is to embed one file within another file.

Requirements:

- The application should accept an image file (e.g., .bmp) along with a text file containing the message to be embedded.
- It should analyze the size of the message file and the data portion of the .bmp image to verify whether the message can fit within the image.
- It should provide an option to embed a magic string, which can be used to identify whether the image contains hidden data.
- The application should provide an option to decrypt (extract) the hidden message.
- The application must be a command-line program, and all options should be passed as command-line arguments.

Flow Chart :-



Error Codes & User Prompts :-

1.User Prompts Function Encoding :-

Case No.	User Prompt
1	Verifying the number of arguments provided
2	Encoding operation detected; validating required arguments
3	Arguments received successfully; initiating validation and parsing
4	Validating .bmp extension for the source image file
5	Checking for file extension in the secret file name
6	Verifying presence of the fourth argument
7	Fourth argument detected; validation in progress
8	Validation successful
9	Fourth argument not provided; assigning default output file name stego.bmp
10	All file names validated successfully
11	Opening required files
12	Encoding process initiated
13	Validating file sizes and checking encoding capacity
14	Sufficient capacity confirmed; copying BMP header to destination file
15	Encoding magic string
16	Encoding secret file extension size
17	Encoding secret file extension
18	Encoding secret file size
19	Encoding secret file data
20	Copying remaining data to destination file
21	Encoding completed; closing all opened files

2.User Prompts Function Decoding :-

Case No.	User Prompt
1	Decoding operation recognized; validating required arguments
2	Validation successful; initiating decoding process
3	Validating second argument
4	— Spare
5	Input .bmp image file validated successfully
6	Opening input .bmp image file
7	— spare
8	Input image file opened successfully
9	Skipping BMP header (first 54 bytes)
10	Extracting magic string
11	Validating magic string and user identification
12	User verified; extracting secret file extension size
13	Extracting secret file extension
14	Generating output file name
15	Opening output file
16	Extracting secret message size
17	Extracting secret message and writing to output file
18	Message extraction completed; closing all opened files
19	Displaying decoded message
20	Decoding completed successfully; refer to output file for the message
21	Output file name not provided; assigning default file name
22	Output file opened successfully
23	Output file name provided; proceeding with further processing

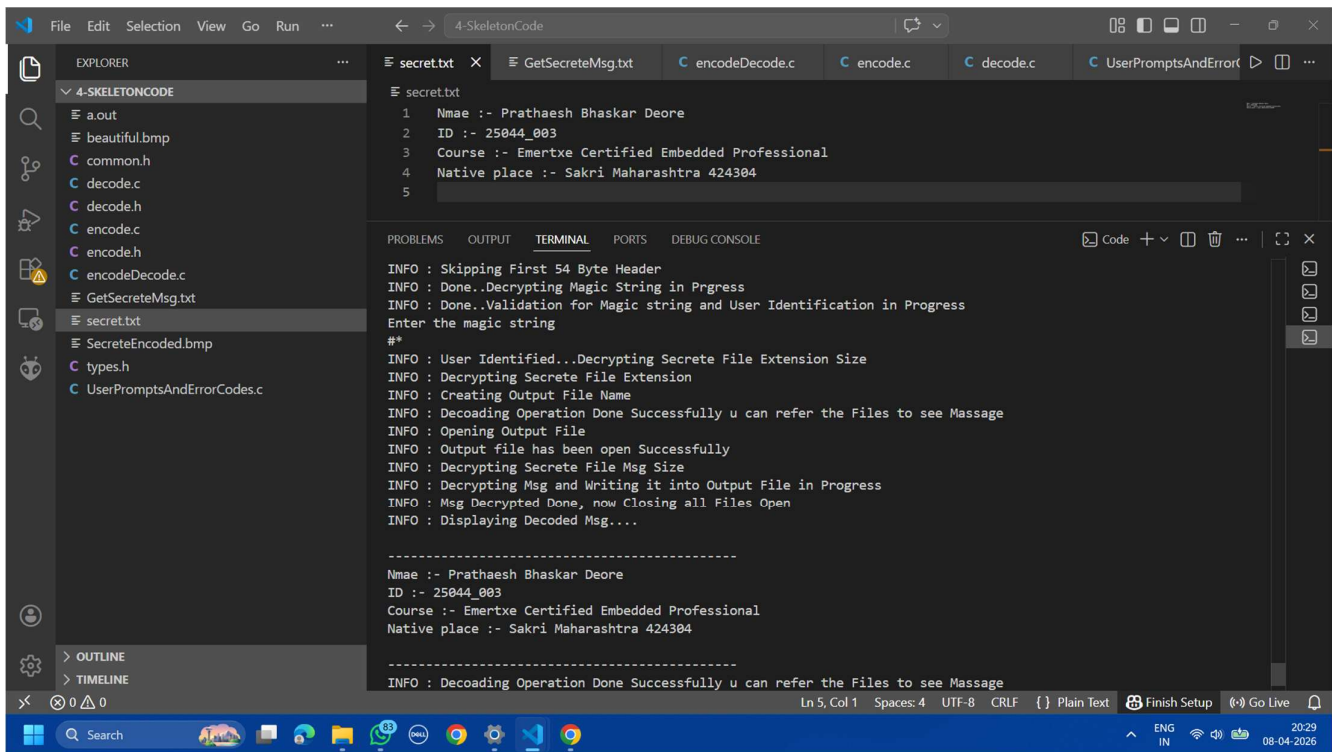
Error Codes Function Encoding :-

Error Code	Error Message
1	Invalid arguments provided
2	Required arguments for encoding are missing or incorrect
3	Invalid source file; only .bmp format is supported
4	Secret file name does not contain a valid extension
5	Output file format not supported; only .bmp files are allowed
6	Error occurred while copying BMP header
7	Error occurred during encoding of magic string
8	Error occurred during encoding of file extension size
9	Error occurred during encoding of secret file extension
10	Error occurred during encoding of secret file size
11	Error occurred during encoding of secret file data
12	Error occurred while copying remaining data from source image to destination image
13	Insufficient capacity in source image to encode complete data

Error Codes Function Decoding

Error Code	Error Message
1	Invalid or insufficient arguments provided for decoding
2	Invalid second argument
3	Error occurred while opening input .bmp file
4	Error occurred during extraction of magic string
5	Magic string verification failed
6	Error occurred during extraction of secret file extension size
7	Error occurred during extraction of secret file extension
8	Error occurred while generating output file name
9	Error occurred while opening output file
10	Error occurred during extraction of secret message
11	Error occurred while writing secret message to output file

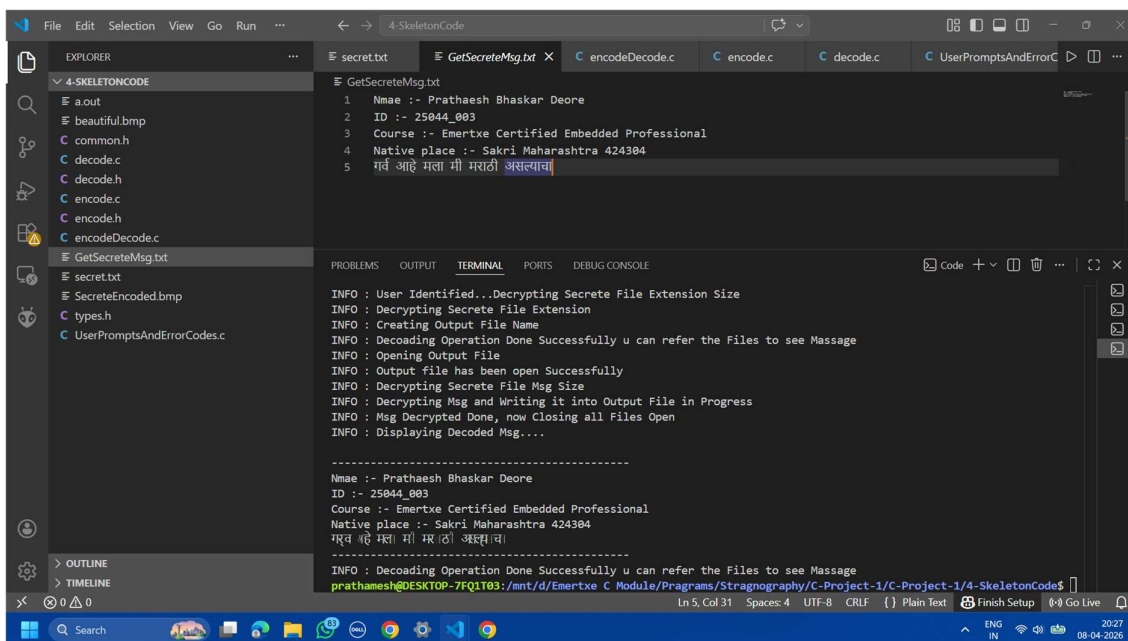
Terminal Outputs :-



```
1 Nmae :- Prathaesh Bhaskar Deore
2 ID :- 25044_003
3 Course :- Emertxe Certified Embedded Professional
4 Native place :- Sakri Maharashtra 424304
5
```

```
INFO : Skipping First 54 Byte Header
INFO : Done..Decrypting Magic String in Progress
INFO : Done..Validation for Magic string and User Identification in Progress
Enter the magic string
**
INFO : User Identified...Decrypting Secrete File Extension Size
INFO : Decrypting Secrete File Extension
INFO : Creating Output File Name
INFO : Decoading Operation Done Successfully u can refer the Files to see Message
INFO : Opening Output File
INFO : Output file has been open Successfully
INFO : Decrypting Secrete File Msg Size
INFO : Decrypting Msg and Writing it into Output File in Progress
INFO : Msg Decrypted Done, now Closing all Files Open
INFO : Displaying Decoded Msg....

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INFO : Decoading Operation Done Successfully u can refer the Files to see Message
```



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```
INFO : User Identified...Decrypting Secrete File Extension Size
INFO : Decrypting Secrete File Extension
INFO : Creating Output File Name
INFO : Decoading Operation Done Successfully u can refer the Files to see Message
INFO : Opening Output File
INFO : Output file has been open Successfully
INFO : Decrypting Secrete File Msg Size
INFO : Decrypting Msg and Writing it into Output File in Progress
INFO : Msg Decrypted Done, now Closing all Files Open
INFO : Displaying Decoded Msg....

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मूव आहे मला मी मराठी असल्याच
-----
INFO : Decoading Operation Done Successfully u can refer the Files to see Message
```

```
File Edit Selection View Go Run ... 4-SkeletonCode
EXPLORER
4-SKELETONCODE
a.out
beautiful.bmp
common.h
decode.c
decode.h
encode.c
encode.h
encodeDecode.c
GetSecreteMsg.txt
secret.txt
SecreteEncoded.bmp
types.h
UserPromptsAndErrorCodes.c
OUTLINE
TIMELINE
0 0 0

secret.txt
1 Nmae :- Prathaes Bhaskar Deore

PROBLEMS OUTPUT TERMINAL PORTS DEBUG CONSOLE
ERROR: Unable to open file secet.txt
Some Error Occured
prathamesh@DESKTOP-7FQ1T03:/mnt/d/Emertxe C Module/Programs/Stragnography/C-Project-1/C-Project-1/4-SkeletonCode$ ./a.out -e beautiful.bmp secret.txt SecreteEncoded.bmp
INFO : Checking Number of arguments Entered
INFO : Encoding Operation Identified Checking Number of arguments Required
INFO : Found ok...Now Validating and Reading Arugements
INFO : checking for .bmp extn in 2nd argument which src img file name
INFO : chk for '.' in 3nd arguement which secreate file name
INFO : Checking whether 4rth argument Present or Not
INFO : 4rth Argument Present..Validating in Prgress...
ERROR : ENCODING : OutPut Source File Provided is not supported, only .bmp Files are Allowed
prathamesh@DESKTOP-7FQ1T03:/mnt/d/Emertxe C Module/Programs/Stragnography/C-Project-1/C-Project-1/4-SkeletonCode$ ./a.out -e beautiful.bmp secret.txt SecreteEncoded.bmp
INFO : Checking Number of arguments Entered
INFO : Encoding Operation Identified Checking Number of arguments Required
INFO : Found ok...Now Validating and Reading Arugements
INFO : checking for .bmp extn in 2nd argument which src img file name
INFO : chk for '.' in 3nd arguement which secreate file name
INFO : Checking whether 4rth argument Present or Not
INFO : 4rth Argument Present..Validating in Prgress...
INFO : Found OK
INFO : All Files Names are Validated and Found OK
INFO : Opening Files In Progress
INFO : ***ENCOADING HAS BEEN STARTING****
fopen: No such file or directory
ERROR: Unable to open file secet.txt
Some Error Occured
prathamesh@DESKTOP-7FQ1T03:/mnt/d/Emertxe C Module/Programs/Stragnography/C-Project-1/C-Project-1/4-SkeletonCode$
```

```
File Edit Selection View Go Run ... 4-SkeletonCode
EXPLORER
4-SKELETONCODE
a.out
beautiful.bmp
common.h
decode.c
decode.h
encode.c
encode.h
encodeDecode.c
GetSecreteMsg.txt
secret.txt
SecreteEncoded.bmp
types.h
UserPromptsAndErrorCodes.c
OUTLINE
TIMELINE
0 0 0

secret.txt
1 Nmae :- Prathaes Bhaskar Deore

PROBLEMS OUTPUT TERMINAL PORTS DEBUG CONSOLE
C decode.c > Do_Decoding(char **, decode_Info *)
1 #include <stdio.h>
2
3 Enter the magic string
4 **
5 INFO : User Identified...Decrypting Secrete File Extension Size
6 INFO : Decrypting Secrete File Extension
7 INFO : Creating Output File Name
8 INFO : Decoading Operation Done Successfully u can refer the Files to see Message
9 INFO : Opening Output File
10 INFO : Output file has been open Successfully
11 INFO : Decrypting Secrete File Msg Size
12 INFO : Decrypting Msg and Writing it into Output File in Progress
13 INFO : Msg Decrypted Done, now Closing all Files Open
14 INFO : Displaying Decoded Msg....
15
16 -----
17 Nmae :- Prathaes Bhaskar Deore
18 ID :- 25044_003
19 Course :- Emertxe Certified Embedded Professional
20 Native place :- Sakri Maharashtra 424304
21 -----
22 INFO : Decoading Operation Done Successfully u can refer the Files to see Message
23 prathamesh@DESKTOP-7FQ1T03:/mnt/d/Emertxe C Module/Programs/Stragnography/C-Project-1/C-Project-1/4-SkeletonCode$
```

